C Future Work

Connections to Geometric Representations. This work, while taking a significant step towards interpreting protein structure prediction, focuses primarily on learning linear representations of sequence information. Future work could establish formal connections between SAE sequence representations and equivariant representations of geometric data (Thomas et al., 2018; Geiger and Smidt, 2022; Lee et al., 2023), particularly investigating how the linear overcomplete basis learned by SAEs map to irreducible representations of geometric transformations in the context of protein structure prediction.

Theoretical Analysis of Matryoshka SAEs. Additionally, theoretical analysis drawing from ordered autoencoding (Rippel et al., 2014; Xu et al., 2021) and information bottleneck methods (Tishby et al., 2000) applied to protein structure prediction could improve our understanding of multi-scale feature learning in biological sequences. Such analysis may provide deeper insights into how hierarchical feature representations emerge and interact within the context of protein language models.

D Matryoshka SAE Implementation Details

The Matryoshka SAE architecture builds upon previous ones through nested groups of increasing size. This progressive constraint naturally encourages the emergence of a feature hierarchy - earlier groups must capture high-level, abstract features to achieve reasonable reconstruction with limited capacity, while later groups can encode more granular details. The key innovation lies in their group-wise decoding process, where each group must reconstruct the input using only its allocated subset of latents.

When processing a protein token embedding $x \in \mathbb{R}^d$ from a PLM, the encoding process follows as:

$$z = \mathrm{BatchTopK}(W_{\mathrm{enc}}x + b_{\mathrm{enc}}) \tag{1}$$

where $W_{\mathrm{enc}} \in \mathbb{R}^{n \times d}$ is the encoder weight matrix, $b_{\mathrm{enc}} \in \mathbb{R}^n$ is the encoder bias, and BatchTopK enforces sparsity by keeping only the $B * K$ highest activations in each batch. This enforces average sparsity while allowing the number of active latents per token to vary.

The decoding process is:

$$\hat{x}^{(m)} = W_{\mathrm{dec}}^{(m)} z_{[1:m]} + b_{\mathrm{dec}} \tag{2}$$

Here, $z_{[1:m]}$ represents the first m components of the latent vector, $W_{\mathrm{dec}}^{(m)} \in \mathbb{R}^{d \times m}$ is the decoder weight matrix restricted to its first m columns, and $b_{\mathrm{dec}} \in \mathbb{R}^d$ is the decoder bias. Each group m must reconstruct the input using only its allocated subset of latents. The total loss combines reconstruction errors across all group sizes M:

$$\mathcal{L}(x) = \sum_{m \in M} \|x - \hat{x}^{(m)}\|_2^2 + \alpha \mathcal{L}_{\mathrm{aux}} \tag{3}$$

where α weights any auxiliary regularization terms. We use the auxiliary loss term from Marks et al. (2024) and Gao et al. (2024) to reduce dead latents.

E Swiss-Prot Evaluation

E.1 Background of Swiss-Prot Evaluation

Swiss-Prot is the gold standard manually curated section of the UniProt Knowledge Base, containing expert-annotated protein sequences with detailed information about their structure, function, modifications, and key biological regions (Poux et al. (2017); Consortium (2024)). Through extensive experimental validation and literature review, Swiss-Prot provides high-quality annotations that span from individual catalytic sites to complete functional domains. Following Simon and Zou (2024), we evaluate whether learned features align with these known biological concepts using a modified F1 score that handles the mismatch between precise feature activations and broader protein annotations present in Swiss-Prot. This approach calculates precision at the amino acid level but recall at the domain level, enabling systematic comparison between learned features and known biological concepts.

E.2 Performance of SAE Models on F1 Concept Evaluation

Model Scale vs Architecture: At 3B scale, architectural differences have minimal impact, with both Matryoshka and TopK identifying 233 concepts and showing only small variations in high-quality feature-concept pairs (2,677 vs 2,461). At 8M scale, architectural differences are more pronounced though still modest, with TopK outperforming Matryoshka (95 vs 72 concepts). However, the scale effect dominates these architectural differences - both 3B variants substantially outperform both 8M variants across all concept types.

Category	Total Concepts	Concepts with F1 > 0.5 (%)			
		3B MatK	3B TopK	8M MatK	8M TopK
Domains	256	193 (75.4%)	195 (76.2%)	50 (19.5%)	72 (28.1%)
Regions	55	19 (34.5%)	20 (36.4%)	10 (18.2%)	13 (23.6%)
Motifs	37	6 (16.2%)	5 (13.5%)	2 (5.4%)	3 (8.1%)
Zinc Fingers	15	5 (33.3%)	5 (33.3%)	2 (13.3%)	2 (13.3%)
Other Features	10	5 (50.0%)	3 (30.0%)	4 (40.0%)	1 (10.0%)
Targeting Peptides	5	5 (100.0%)	5 (100.0%)	4 (80.0%)	4 (80.0%)
Total	476	233 (48.9%)	233 (48.9%)	72 (15.1%)	95 (20.0%)

Table 1: Concept coverage comparison across model scales and architectures. We report the number (percentage) of concepts with F1 scores exceeding 0.5 for each model variant, broken down by concept category.

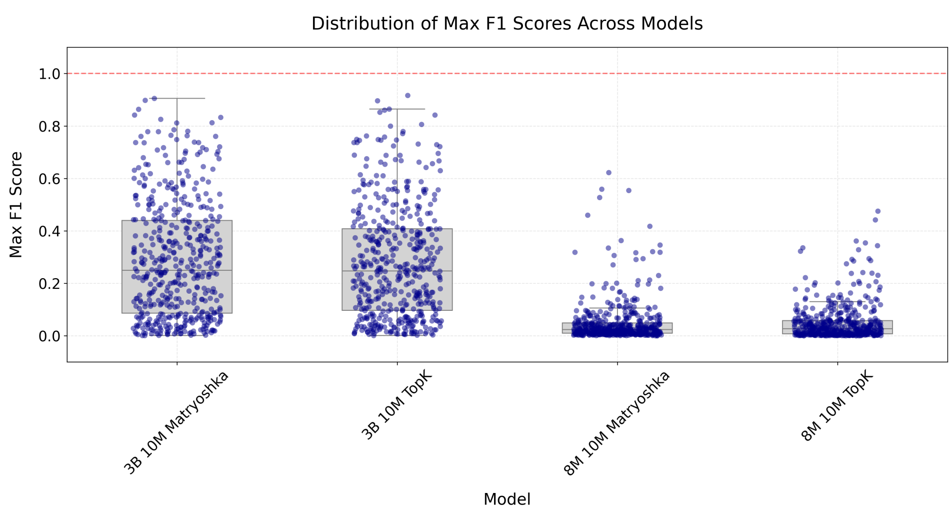

Figure 5: Distribution of highest F1 scores achieved for each concept across models.

The boxplot distribution reveals a clear separation between model scales (5. The 3B models show broader distributions with medians around 0.4 and numerous concepts achieving scores above 0.6. In contrast, 8M models show compressed distributions with medians below 0.1, indicating substantially weaker concept capture across the board.

The comparison heatmaps demonstrate relative performance between model variants by comparing their best-performing features for each concept. The left heatmap shows the number of concepts where each row model achieves a higher F1 score than the column model, while the right heatmap shows the average F1 score difference (Figure 6). The 3B models

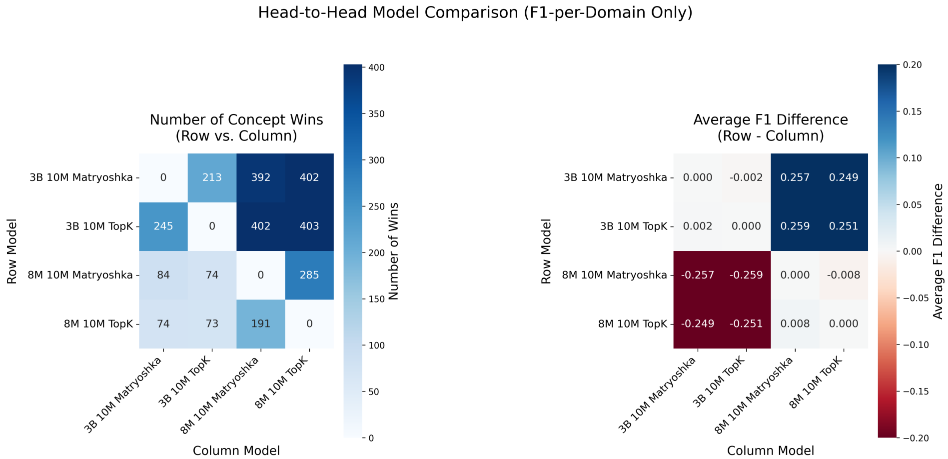

Figure 6: Left: Number of concepts where the row model achieves a higher maximum F1 score for a given concept than the column model. Right: Average score difference of the highest F1 scores for a given concept between models (row minus column) across all concepts. Each cell compares the best-performing feature for each concept between model pairs. Darker blue indicates stronger performance advantage for the row model.

outperform 8M variants on approximately 400 concepts, with an average F1 improvement of 0.25. Within each scale, architectural variants show minimal differences (average F1 difference < 0.01), suggesting model scale rather than architecture drives concept capture ability.